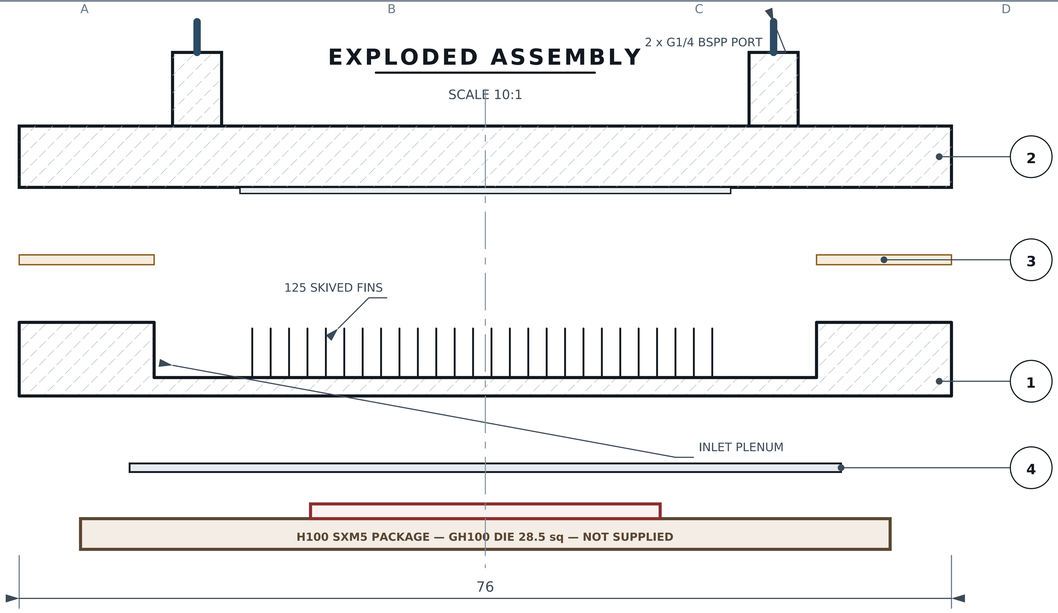
EXPLODED ASSEMBLY

SCALE 10:1



ASSEMBLED HEIGHT 11 mm + 6 PORT BOSS = 17 mm

BILL OF MATERIALS — ASSEMBLY CP-100

ITEM	QTY	PART No.	DESCRIPTION	MATERIAL	NOTE
1	1	CP-101	COLD PLATE BODY, SKIVED FIN	Cu C10200 / C1020	sheet 2
2	1	CP-102	COVER / MANIFOLD, 2 PORTS	Cu C10200 / C1020	sheet 3
3	1	CP-103	BRAZE PREFORM, PERIMETER RING	BCuP-5 (Ag 15%)	0.10 foil
4	1	CP-104	THERMAL INTERFACE, BARE DIE	PTM, k >= 13 W/mK	60 um BLT
5	4	CP-201	SPRING SCREW + BELLEVILLE STACK	A2-70 / 17-7PH	sheet 5
6	2	CP-202	QUICK DISCONNECT, DRIP-FREE	Ni-plated brass	loop side

RESISTANCE BUDGET — PG25, 1.5 L/min

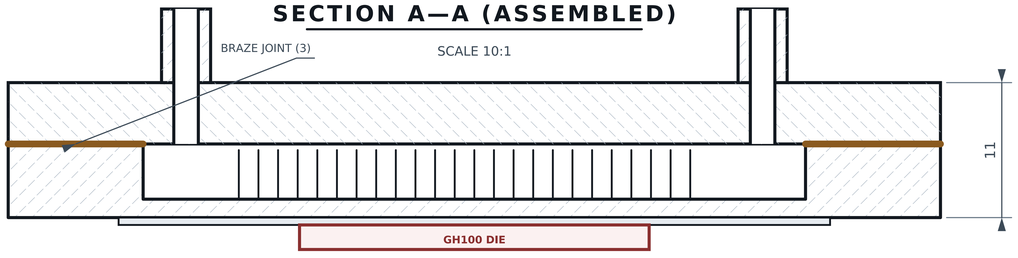
PATH	K/W	dT @ 700 W	SHARE
coolant (convection + sensible)	**0.0141**	9.9 K	48 %
TIM, bare die 60 um k=13	0.0057	4.0 K	19 %
spreading, die to fin field	0.0050	3.5 K	17 %
base 1.5 mm Cu	0.0047	3.3 K	16 %
TOTAL, die to coolant inlet	**0.0295**	**20.7 K**	**100 %**

DIE TEMPERATURE — SAME PLATE, 1.5 L/min

COOLANT IN	H100 700 W	B200 1000 W	GB200 1200 W
32 C	53 C	62 C	67 C
40 C	61 C	70 C	75 C
45 C	**66 C**	**75 C**	**80 C**
50 C	71 C	80 C	85 C

SECTION A—A (ASSEMBLED)

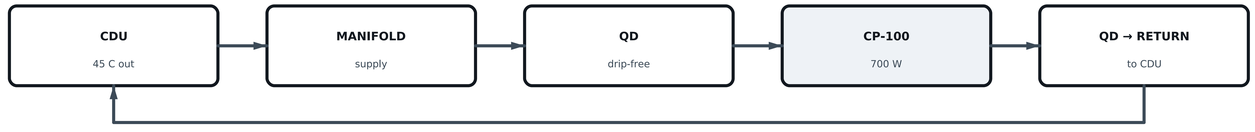
SCALE 10:1



Coolant enters left plenum, crosses 125 channels, leaves right plenum.

LOOP SCHEMATIC

ONE GPU SHOWN



per GPU: 1.5 L/min PG25, 6.1 kPa at the plate, 7.0 K coolant rise

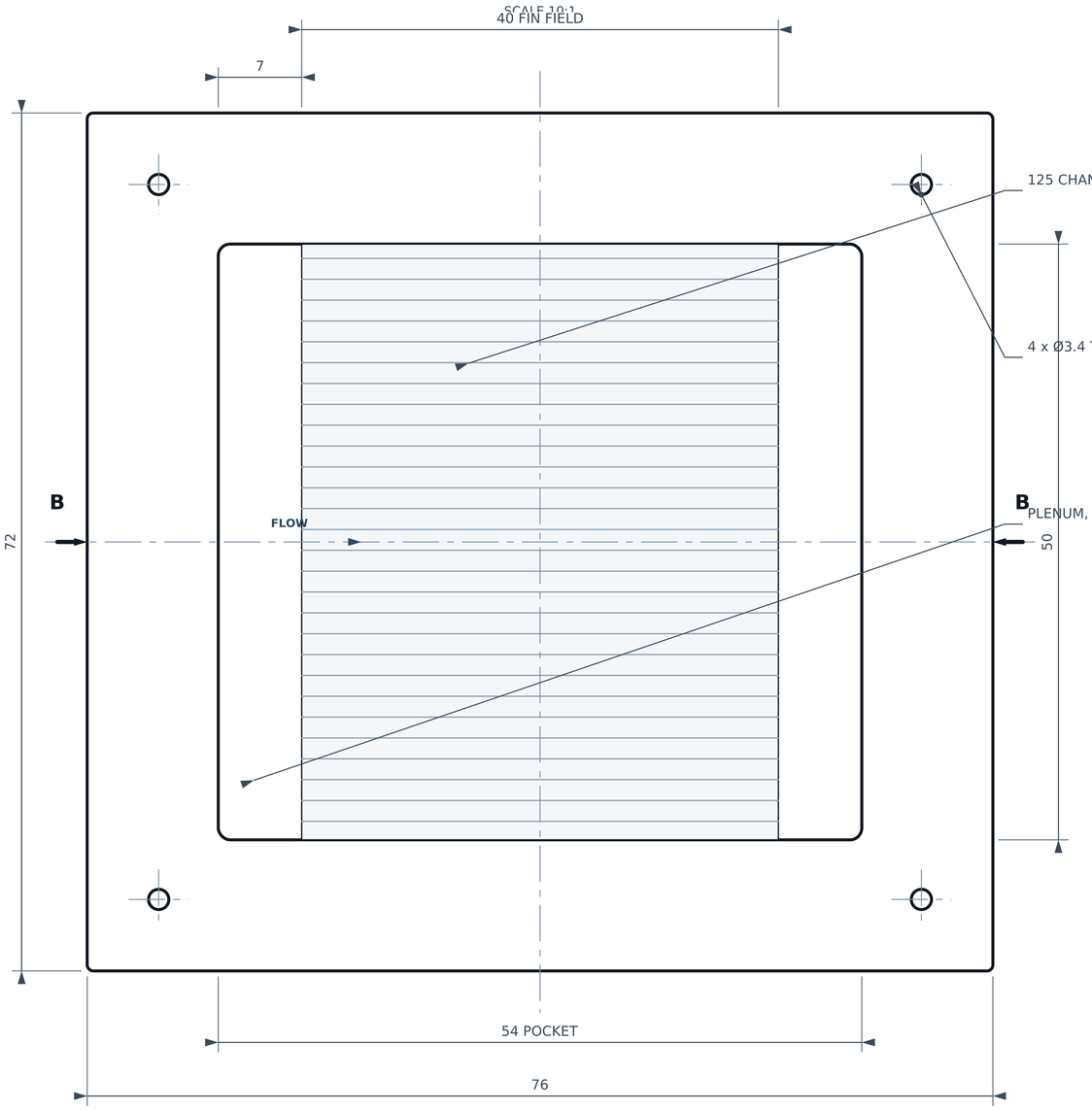
GENERAL NOTES

1. Interpret per ISO 128. Tolerances ISO 2768-mK unless stated. Dimensions in mm.
2. EVERY WETTED PART IS COPPER OR STAINLESS. No aluminium anywhere in the loop — a Cu/Al couple in glycol corrodes the aluminium out within months.
3. Cover is vacuum brazed to the body: no elastomer inside the loop, no serviceable joint. Replace, do not open.
4. SXM5 interface dimensions and mounting preload on sheet 5 are ASSUMED. Confirm against NVIDIA's mechanical specification before manufacture.

INDIA-VC THERMAL HARDWARE	ISO 128 / ISO 2768-mK
TITLE: **D2C COLD PLATE — GENERAL ASSEMBLY** — H100 SXM5 / 700 W / Cu C10200, VACUUM BRAZED / PG25 1.5 L/min	SCALE: 10:1 (THIRD ANGLE PROJECTION) · UNITS: mm
DRAWING No.: **CP-100**	REV: **A** · SHEET: **1 / 6**
DRAWN: AI-ASSISTED DRAFT · CHECKED: — PENDING — · MATERIAL: SEE BOM · FINISH: SEE NOTES · DATE: 2026-09-10	

PLAN — FIN SIDE UP

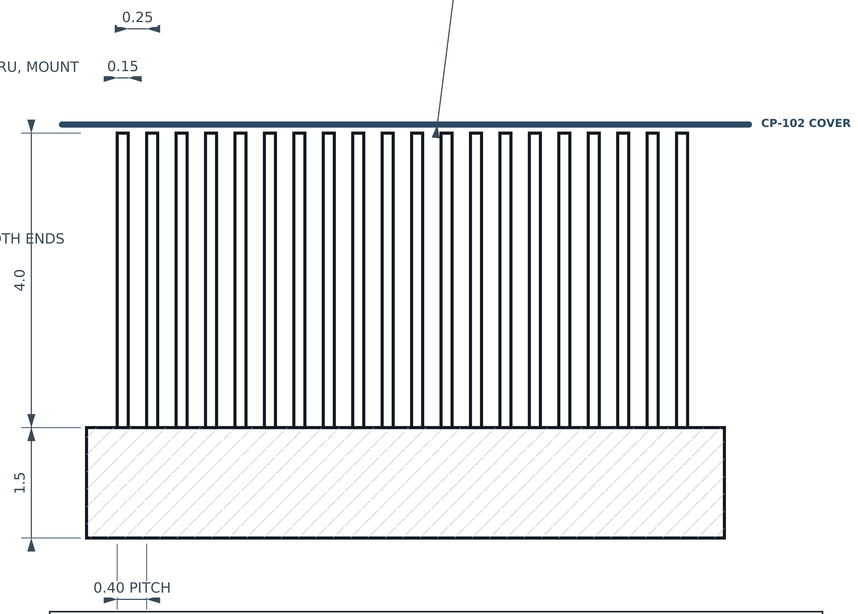
SCALE 10:1



DETAIL E — FIN ROOT

TIP CLEARANCE 0.05 MAX — NOT BRAZED

SCALE 60:1



SECTION B—B (ALONG FLOW)

SCALE 10:1

NOTES

1. MATL: OXYGEN-FREE copper C10200 / JIS C1020.
2. Fins skived from solid. No brazed or bonded fin.
3. Braze land (perimeter, 8 wide) flat 0.02, Ra 0.8.
4. Die-side face flat 0.015, Ra 0.4, no burr.
5. Deburr channel ends. A raised burr blocks a channel.
6. Flush and dry before braze. No cutting fluid in the fins.
7. Handle fin field with a cover plate. Bent fins are scrap.

CHANNEL PERFORMANCE (PG25)	
channels	125
hydraulic diameter	0.471 mm
velocity @ 1.5 L/min	0.20 m/s
Reynolds	71 laminar
h	5300 W/m2K
fin efficiency	0.55
pressure drop	6.1 kPa

Aspect ratio 16:1. Skiving is the only economical way to cut 0.15 mm fins this tall in copper.

CRITICAL CHARACTERISTICS			
FEATURE	**NOMINAL**	**TOL**	**CRITICAL TO**
Base thickness under die	1.5	+0.05 / -0.00	conduction, 3.3 K
Fin height	4.0	±0.05	flow area / dP
Channel width	0.25	±0.02	dP goes as w^-2
Die-side flatness	—	0.015	TIM bond line
Braze land flatness	—	0.02	hermetic joint
Channel blockage	zero	100 % flow test	one blocked channel = hot spot

INDIA-VC THERMAL HARDWARE — ISO 128 / ISO 2768-mK

TITLE: **COLD PLATE BODY**
Cu C10200 (JIS C1020) / 76 x 72 x 6.0 / 125 SKIVED FINS

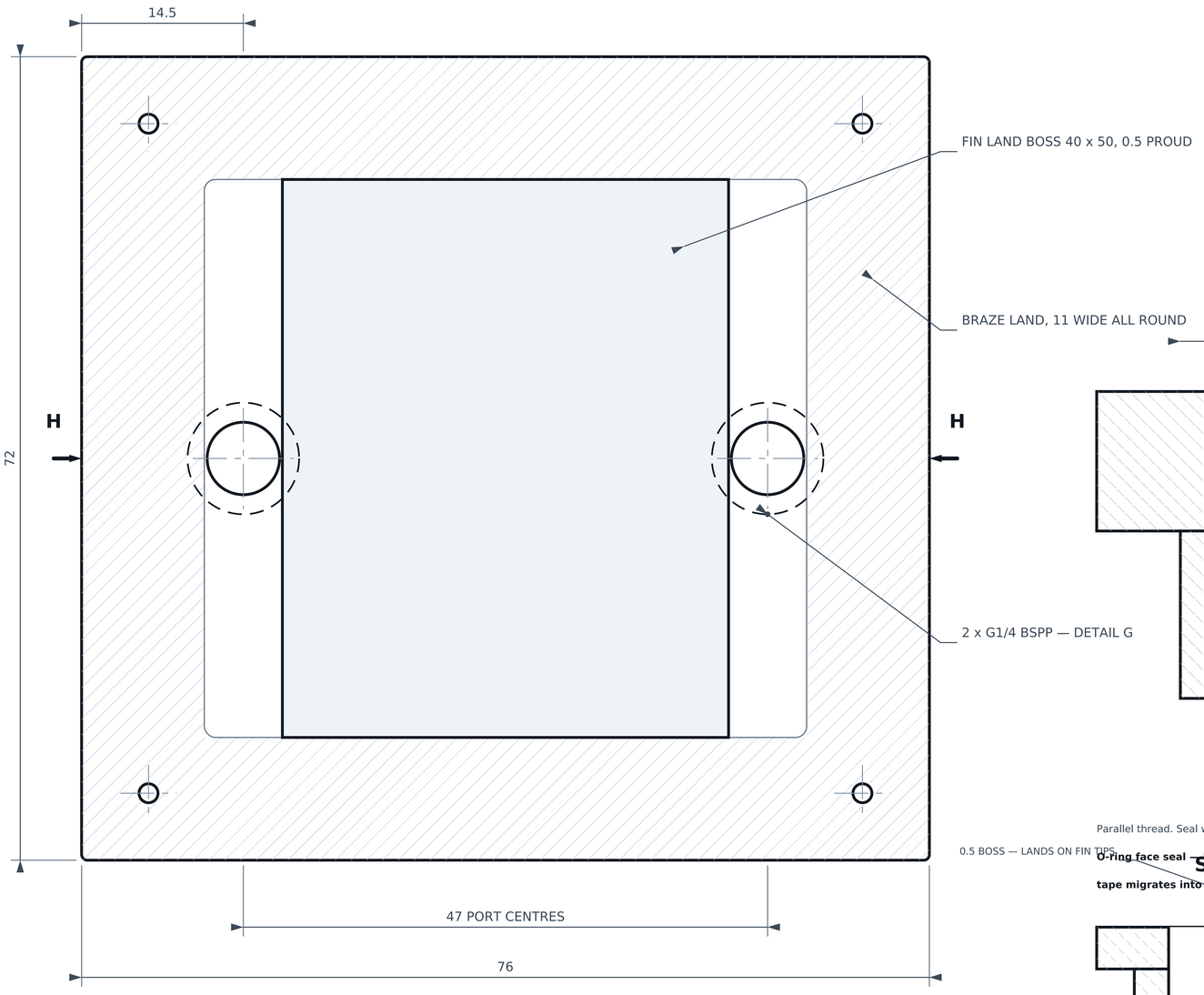
SCALE: **10:1** — UNITS: **mm** — THIRD ANGLE PROJECTION

DRAWING No. **CP-101** — REV **A** — SHEET **2 / 6**

DRAWN	CHECKED	MATERIAL	FINISH	DATE
AI-ASSISTED DRAFT	— PENDING —	SEE BOM	SEE NOTES	2026-09-10

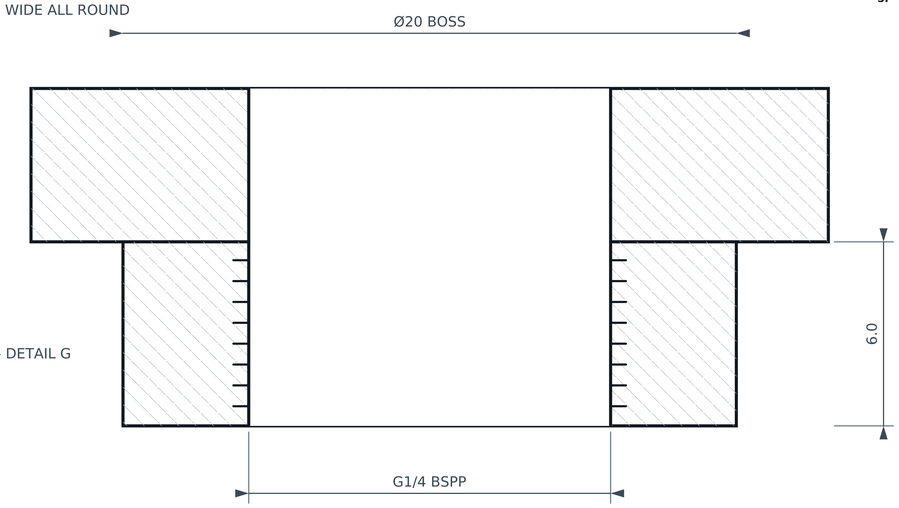
PLAN — UNDERSIDE (WETTED FACE)

SCALE 10:1



DETAIL G — PORT

SCALE 25:1



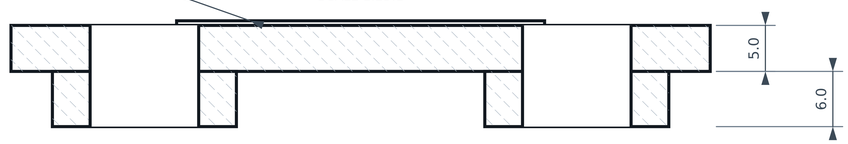
Parallel thread. Seal with a bonded washer or

0.5 BOSS — LANDS ON FIN TIPS. **O-ring face seal — NOT with PTFE tape: shredded**

tape migrates into the 0.25 mm channels.

SECTION H—H (WETTED FACE UP)

SCALE 1.25:1



Ports open straight into the plenums; no internal turn.

NOTES

1. MATL: OXYGEN-FREE copper C10200 / JIS C1020.
2. Braze land flat 0.02, Ra 0.8, matched to CP-101.
3. Fin land boss 0.5 proud, flat 0.02 — it seals the fin field against bypass flow.
4. Deburr port bores fully. Any chip ends up in a channel.
5. Do not braze the fin land. Mechanical contact only.

INSPECTION

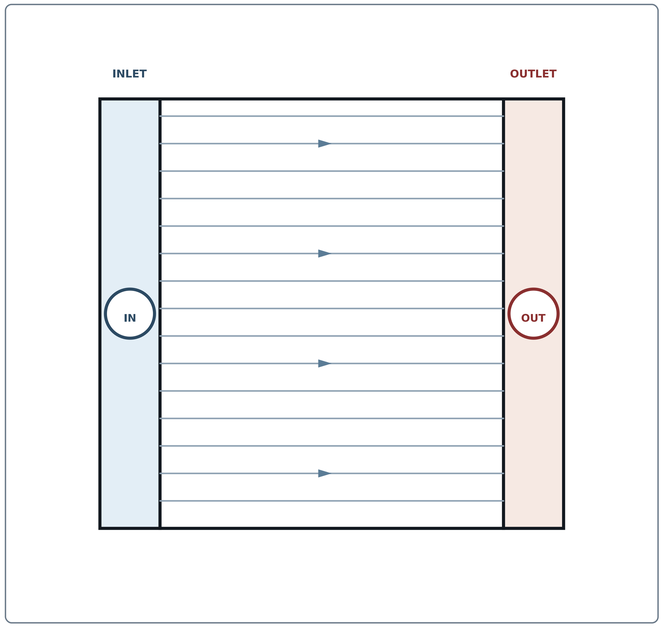
CHECK	REQUIREMENT	METHOD
Braze land flatness	0.02 max	CMM / optical flat
Fin land boss height	0.50 ± 0.02	height gauge
Port thread	G1/4 BSPP, gauge to BS 2779	ring gauge
Cleanliness	no chips, no cutting fluid	borescope + flush

INDIA-VC THERMAL HARDWARE — ISO 128 / ISO 2768-mK

TITLE: **COVER / MANIFOLD** — Cu C10200 (JIS C1020) / 76 x 72 x 5.0 / 2 x G1/4 BSPP	SCALE: **10:1** THIRD ANGLE PROJECTION	UNITS: **mm**	
DRAWING No.: **CP-102**	REV: **A**	SHEET: **3 / 6**	
DRAWN: AI-ASSISTED DRAFT	CHECKED: — PENDING —	MATERIAL: SEE BOM	FINISH: SEE NOTES
DATE: 2026-09-10			

FLOW PATH

SCALE 1.20:1



125 PARALLEL CHANNELS — 0.25 W x 4.0 H x 40 LONG

Single pass, straight. No U-turn, no serpentine:

at 0.25 mm a turn costs more than the whole fin field.

DESIGN POINT — 1.5 L/min, 700 W

PARAMETER	WATER	PG25
velocity in channel	0.20 m/s	**0.20 m/s**
Reynolds number	170	**71**
heat transfer coefficient	7641 W/m2K	**5300 W/m2K**
fin efficiency	0.47	**0.55**
coolant temperature rise	6.8 K	**7.0 K**
pressure drop, plate only	2.5 kPa	**6.1 kPa**

ROBUSTNESS — CONSTANT FLOW

CHANNEL BLOCKAGE	OPEN	VELOCITY	R total	DIE @45 C	dP
all clear	**125**	**0.20 m/s**	**0.0295**	**65.7 C**	**6.1 kPa**
5 % blocked	118	0.21 m/s	0.0299	66.0 C	6.5 kPa
10 % blocked	112	0.22 m/s	0.0303	66.2 C	6.9 kPa
20 % blocked	100	0.25 m/s	0.0313	66.9 C	7.7 kPa

READING THESE NUMBERS

- PG25 is the design fluid. Water is shown only to bound the gain from inhibited water where freeze protection is not required — about 5 % on total resistance.
- A SINGLE blocked channel is not a thermal event. The 1.5 mm copper base carries its ~10 W sideways 0.4 mm to the neighbours for roughly 0.1 K.
- Progressive fouling across many channels is the real risk, and it shows up first as rising pressure drop, not as a rising die temperature.
- So instrument the loop for DIFFERENTIAL PRESSURE, not only temperature, and filter to 50 um at the CDU. Service trigger: dP 20 % above the commissioning baseline at the same flow.

FLOW SENSITIVITY — PG25, 700 W

FLOW	R total	DIE @45 C	RISE	dP
0.75 L/min	0.0373	71.1 C	14.0 K	3.1 kPa
1.00 L/min	0.0333	68.3 C	10.5 K	4.1 kPa
1.50 L/min	**0.0295**	**65.7 C**	**7.0 K**	**6.1 kPa**
2.00 L/min	0.0278	64.4 C	5.2 K	8.2 kPa
3.00 L/min	0.0261	63.3 C	3.5 K	12.3 kPa

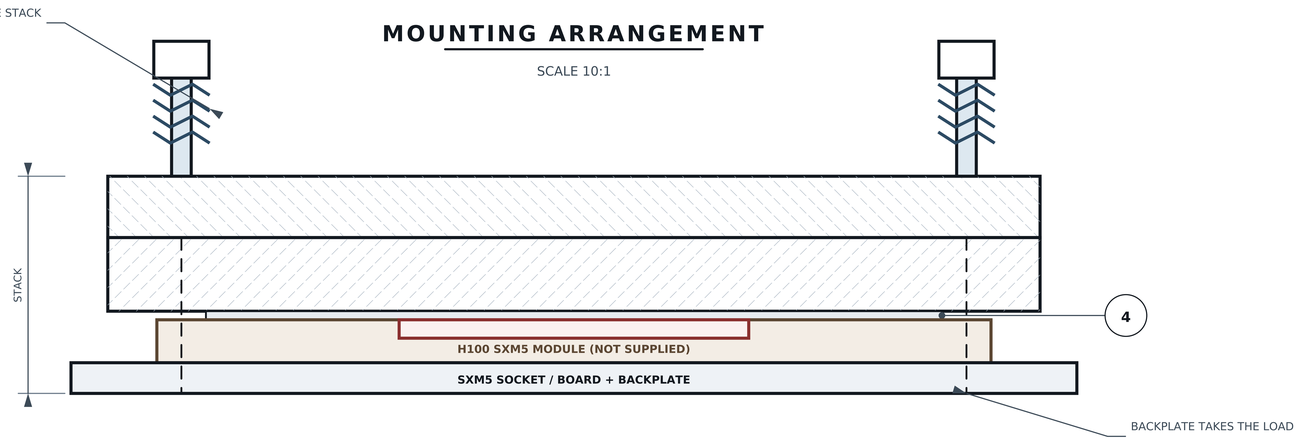
Below 1.0 L/min the plate loses ground quickly. Above 2.0 L/min it buys almost nothing — spend the pump head elsewhere in the loop.

INDIA-VC THERMAL HARDWARE				ISO 128 / ISO 2768-mK
TITLE: **THERMAL-HYDRAULIC SPECIFICATION** — H100 SXM5 700 W / PG25 / DESIGN POINT AND SENSITIVITIES			SCALE: **NONE** — THIRD ANGLE PROJECTION	UNITS: **mm**
DRAWING No. **CP-103**			REV **A**	SHEET **4 / 6**
DRAWN: AI-ASSISTED DRAFT	CHECKED: — PENDING —	MATERIAL: SEE BOM	FINISH: SEE NOTES	DATE: 2026-09-10

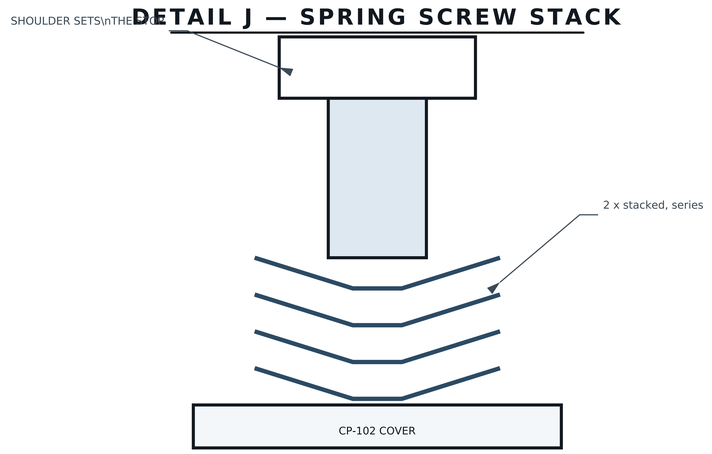
ASSUMED INTERFACE — CONFIRM EVERY DIMENSION AND THE PRELOAD AGAINST NVIDIA'S SXM5 MECHANICAL SPECIFICATION BEFORE MANUFACTURE

MOUNTING ARRANGEMENT

SCALE 10:1



DETAIL J — SPRING SCREW STACK



The shoulder, not the torque wrench, sets the preload.

Torque-only assembly on a bare die cracks silicon.

MOUNTING PARAMETERS

PARAMETER	VALUE	SOURCE
Total preload on the package	**TO BE TAKEN FROM NVIDIA SPEC**	**NOT SET HERE**
Screws	4 x M3 x 0.5, shouldered	this drawing
Spring element	2 x stacked, series	this drawing
Tightening	cross pattern, 3 passes to shoulder	this drawing
Cold plate flatness, die face	0.015 max	CP-101
Parallelism, plate to package	0.05 max	this drawing
TIM	PTM, k >= 13 W/mK, 60 um BLT	CP-104

WHY THIS SHEET IS NOT FINISHED

- An SXM5 module is a bare die on a socketed carrier. The socket, the load frame and the allowable preload are defined by NVIDIA, not by the cold plate designer.
- Too much preload cracks the die or the substrate. Too little starves the TIM bond line and the 4.0 K TIM budget on sheet 1 becomes 15 K or worse.
- Obtain the SXM5 mechanical and thermal design guide, then set the preload, the screw shoulder height and the Belleville stack to match it. Everything else on these six sheets is independent of that number.
- Do not substitute a torque specification for a shoulder stop. Torque on an M3 screw scatters by +/- 30 % with friction alone.

MOUNTING RISK REGISTER

RISK	MECHANISM	CONTROL
Die crack	**preload above the package rating, or uneven load**	**shoulder stop + cross tightening**
High TIM resistance	insufficient preload, or air in the bond line	preload to spec, controlled dispense
TIM pump-out	thermal cycling with a soft grease	use a phase-change material
Plate rocking	flatness or parallelism out of tolerance	0.015 / 0.05 per this sheet

INDIA-VC THERMAL HARDWARE — ISO 128 / ISO 2768-mK

TITLE: **SXM5 MOUNTING INTERFACE** — LOAD FRAME, PRELOAD AND TIM / DIMENSIONS ASSUMED — SEE BANNER	SCALE: **10:1** THIRD ANGLE PROJECTION	UNITS: **mm**	
DRAWING No.: **CP-200**	REV: **A**	SHEET: **5 / 6**	
DRAWN: AI-ASSISTED DRAFT	CHECKED: — PENDING —	MATERIAL: SEE BOM	FINISH: SEE NOTES	DATE: 2026-09-10

MANUFACTURE AND TEST SEQUENCE

IN ORDER

1 — MACHINE

CP-101 body and CP-102 cover. Pocket, plenums, ports, mount holes.

2 — SKIVE FINS — one pass per fin

125 fins, 0.15 thk x 4.0 tall @ 0.40 pitch.

3 — DEBURR + CLEAN — no cutting fluid

Ultrasonic in alkaline, DI rinse, dry. Borescope every channel.

4 — BRAZE — fin tips NOT brazed

Vacuum or dry N2/H2. BCuP-5 preform on the perimeter land ONLY.

5 — PROOF PRESSURE — typically 15 bar

Hydrostatic 3 x working, 5 min, no permanent set.

6 — LEAK TEST — < 1e-8 Pa.m3/s

Helium mass spectrometer under vacuum.

7 — FLOW TEST — 6.1 kPa nom

Measure dP at 1.5 L/min. Record as the commissioning baseline.

8 — THERMAL TEST — first article + AQL

Die simulator at 700 W. Verify R <= 0.032 K/W.

9 — FLUSH + DRY — ship dry, capped

DI flush, N2 purge, cap both ports.

10 — INSTALL — air blankets fins

Mount per CP-200. Fill, vent, run 30 min, re-vent.

COOLANT SPECIFICATION

PROPERTY	SPECIFICATION	WHY
Fluid	**PG25, 25 % propylene glycol**	**freeze + biocide**
Inhibitor package	phosphate/azole, Cu-rated	protects copper
pH	8.0 - 9.5	copper passivity
Conductivity	< 50 uS/cm	galvanic / leak sensing
Particulate	filtered 50 um at the CDU	0.25 mm channels
Supply temperature	45 C nominal, 50 C max	warm-water loop
Flow per plate	1.5 L/min	design point
Change interval	per fluid supplier, test annually	inhibitor depletes

WETTED MATERIALS — GALVANIC COMPATIBILITY

MATERIAL	USE	VERDICT
Copper C10200, brass, bronze	plate, fittings	**ALLOWED**
Stainless 304 / 316	tubing, QD bodies	**ALLOWED**
EPDM	hose, external seals	**ALLOWED**
ALUMINIUM, any alloy	**anywhere wetted**	**PROHIBITED**
Zinc, galvanised steel	anywhere wetted	**PROHIBITED**
Nitrile, PTFE tape	seals	not recommended

Copper is strongly cathodic. One aluminium fitting in a copper loop corrodes

through in months and the debris then blocks the 0.25 mm channels.

COMMISSIONING

- Fill slowly from the lowest point. Vent at the highest point until clear.
- Run 30 minutes, re-vent, then record dP at 1.5 L/min as the baseline for this plate.
- Trend dP, not only temperature. Fouling shows in dP long before the die gets hot.
- Service trigger: dP 20 % above the commissioning baseline at the same flow.

ACCEPTANCE

TEST	LIMIT	APPLIED TO
Helium leak	**< 1e-8 Pa.m3/s**	**100 %**
Proof pressure	3 x working, no set	100 %
dP at 1.5 L/min	6.1 kPa +20 % / -30 %	100 %
Thermal R, die simulator	<= 0.032 K/W at 700 W	first article + AQL

INDIA-VC THERMAL HARDWARE — ISO 128 / ISO 2768-mK

TITLE: **PROCESS, TEST & COOLANT SPECIFICATION**

CP-100 COLD PLATE / MANDATORY SEQUENCE AND LOOP CHEMISTRY

SCALE: NONE — UNITS: mm — THIRD ANGLE PROJECTION

DRAWING No.: **CP-300** — REV: **A** — SHEET: **6 / 6**

DRAWN	CHECKED	MATERIAL	FINISH	DATE
AI-ASSISTED DRAFT	— PENDING —	SEE BOM	SEE NOTES	2026-09-10